REIHAN APRIANDI

Forestry | GIS & Remote Sensing | Environmental Data
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PROFILE

Final-year Forestry undergraduate at Universitas Riau with field and professional experience in land compliance supervision, above-ground biomass surveys, land cover validation, and carbon stock assessment. I use GIS, remote sensing, and Python to measure land cover change, surface heat, satellite observation coverage, and biodiversity data quality. I am open to entry-level GIS analysis, remote sensing, and environmental data roles.

EDUCATION

Bachelor of Science in Forestry Aug 2022 - Present
Universitas Riau, Faculty of Agriculture, Pekanbaru, Indonesia

- Final-year student, 9th semester; GPA 3.52/4.00.
- Focus: forestry, spatial analysis, remote sensing, and climate applications.

PROJECTS

Flood Coverage - Satellite Observation Coverage for Flood Monitoring Nov 2026
Solo project | Python, STAC, Sentinel-1, GloFAS | Public source

- Measured whether Sentinel-1 observes a flood while it is rising, using one year of acquisition metadata across four basins in Italy, Germany, and Indonesia (694 overpasses, Nov 2025 - Nov 2026).
- Found that only 3-21% of 6-hour rise windows were observed; Jakarta forecast peaks were checkable within one day 28% of the time, against 91% on the Po.
- Built a dependency-free Python pipeline over STAC and Open-Meteo, with an interactive web front end and exportable figures.

BioSift - Species Occurrence Data Quality App 2026
Solo project | Python, Streamlit, pygbif, scikit-learn | Submitted to the 2026 GBIF Ebbe Nielsen Challenge

- Built a Streamlit app that fetches GBIF occurrence records, runs ten automated quality checks, and scores each dataset from 0 to 100 for reliability.
- Added DBSCAN spatial outlier detection, kernel density previews, multi-species comparison, and Darwin Core Archive export.
- Deployed a public demo and documented the method for each check.

Urban Surface Temperature Mapping - Pekanbaru 2026
Solo project | Python, Landsat 8/9, Google Earth Engine, rasterio, NumPy

- Mapped land surface temperature across Pekanbaru from Landsat thermal data. A medoid composite and regression fill raised clear coverage from 43.8% to 79.9%, with a per-pixel uncertainty layer.
- Ranked three intervention zones across 125 grid cells; mean surface temperature was 35.9 C (range 27.9-43.7 C), combined with NDVI change.

Urban Green Space Change - Kota Pekanbaru, 2019-2025 2026
Solo project | Python, Sentinel-2, Google Earth Engine, ESA WorldCover

- Measured NDVI change over 62,110 ha of Kota Pekanbaru at 20 m; 6,356 ha lost vegetation, 14,318 ha gained, for a net +7,962 ha.
- Attributed change to ESA WorldCover classes and cross-checked the spatial pattern against Landsat ($r = 0.45$ across 182 cells).

Species Distribution Model Comparison - Four Algorithms 2026
Solo project | Python, MaxEnt, scikit-learn, XGBoost, GBIF

- Compared MaxEnt, random forest, RBF SVM, and XGBoost on the same GBIF records, background sample, and folds for three species.
- Spatial block cross-validation cut AUC by 0.10-0.17 against random folds. MaxEnt reached 0.725 AUC on the smallest case (91 presences), where the other models ranged from 0.492 to 0.654.

Palm Tree Counting - UAV Orthophotos 2026
Solo project | Python, YOLOv8, GDAL, Google Colab, QGIS | Public source

- Counted 24,966 oil palm trees over 310.8 ha from UAV orthophotos, with georeferenced points and crown sizes.
- Merged orthophotos before inference and applied global spatial deduplication; production count bias was +/-8%, with 92% agreement against an independent digitizing pass.

Brovey Fusion - HLS and PlanetScope 2026
Independent analysis | Python, rasterio, NumPy

- Fused a 30 m HLS composite with 3 m PlanetScope imagery using Brovey; measured detail transfer, radiometric bias, and spectral index retention across four 9 km windows for 2016 and 2025.
- Found that panchromatic detail did not follow the class boundaries in the thesis map, while NDVI and spectral angle were unchanged.

LULC Composition PCA - Giam Siak Kecil 2026

Independent analysis | Python, rasterio, NumPy, PCA

- Analyzed five land cover classes across 2000, 2016, and 2025 as a 15-part composition on 990 m blocks using centered log-ratio PCA.
- Retained four components; rebuilt land cover maps agreed with the classified maps for 93.25-97.13% of pixels after generalization.

PROFESSIONAL EXPERIENCE

Site Compliance Supervisor Aug 2026 - Sep 2026

Ministry of Agriculture of the Republic of Indonesia | Contract | Teluk Meranti, Riau, Indonesia | On-site

- Supervised excavation work on site so it stayed inside designated plot boundaries and did not encroach on neighboring areas.
- Directed field workers on boundary stake (patok) placement and guided excavator operators on irrigation channel, embankment (tanggul), and paddy field excavation.
- Coordinated stake placement across multiple fields to define plot limits before work began.

Assistant Surveyor - Above-Ground Biomass (AGB) Peatland Study Jun 2026 - Jul 2026

PT Hatfield Indonesia | Freelance | Riau, Indonesia

- Collected forest plot measurements (DBH, height, species identification) for an above-ground biomass and carbon stock assessment across tropical peatland.
- Recorded field data with standardized protocols and worked with senior surveyors to apply allometric equations and support carbon stock calculations.

Assistant Analyst (Intern) Jan 2025 - Jun 2025

RECOFTC - The Center for People and Forests | Hybrid | Indonesia

- Digitized land cover and land use features from satellite imagery in Collect Earth Online, ArcGIS, and QGIS using predefined classification protocols.
- Validated land cover and land use data against field reference data with confusion matrix evaluation; achieved approximately 90% accuracy.
- Conducted field checks to compare satellite interpretations with ground conditions and maintained methodological consistency with the project team.

Capstone Project Participant - Student Stress Analysis Oct 2025

Hacktiv8 x IBM SkillsBuild | Online

- Cleaned and analyzed survey data in Python (Pandas, Matplotlib) and produced a dashboard report on stress drivers for non-technical stakeholders.

Forest Management Field Training Participant Jul 2024

HPGW, IPB University | West Java, Indonesia

- Completed field practicum in community forestry, forest inventory, nursery management, soil conservation, non-timber forest products, block area mapping, and in-situ conservation.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Expert Staff, Commission 4 Jan 2024 - Dec 2024

Student Representative Council (DPM), Universitas Riau

- Contributed to institutional policy discussions and student governance within the campus legislative body.

Expert Staff, Fundraising & Business Division May 2023 - Sep 2024

Forestry Student Club (FSC), Universitas Riau

- Coordinated fundraising activities and managed business initiatives to support club operations and academic programs.

SKILLS

- **GIS & Remote Sensing:** ArcGIS, QGIS (Advanced), Google Earth Engine, GeoDa, GRASS GIS, Collect Earth Online.
- **Field & Compliance:** Boundary verification, site supervision, encroachment prevention, permit compliance, AGB and carbon stock surveys.
- **Data Analysis:** Python (Pandas, NumPy, Matplotlib), R/RStudio, SQL, exploratory data analysis, data visualization.
- **Remote Sensing Methods:** Land cover change, NDVI, LST, image compositing, gap filling, accuracy assessment, confusion matrix, spatial cross-validation.
- **Web & Dashboards:** Streamlit, Leaflet.js, Folium, HTML/CSS.
- **Machine Learning:** Supervised classification, scikit-learn, DBSCAN, YOLOv8.
- **Tools:** Microsoft Office, Agisoft Metashape, OpenDroneMap.
- **Language:** Indonesian, English (B1 Intermediate).

CERTIFICATIONS

- **Remote Sensing & GIS:** Advanced QGIS (Spatial Thoughts, Apr-May 2026); GIS for Climate Action (Esri, Mar 2026); Visualizing Land Cover and Land Use Change with NASA Satellite Imagery (NASA ARSET, Feb 2026); Monitoring Water Quality in Lakes and Coastal Regions Using STREAM (NASA ARSET, Feb 2026); Hyperspectral Data for Land and Coastal Systems (NASA ARSET, Jan 2026); Advanced NASA Earth Observations and Tools for Active Fire, Smoke, and Post-Fire Monitoring (NASA ARSET, Jan 2026); Fundamentals of Remote Sensing (NASA ARSET, Dec 2025).
- **Data Analytics & AI:** Google Data Analytics (Coursera/Google, Oct 2025); Oracle Cloud Infrastructure 2025 Certified Data Science Professional (Oracle University, Oct 2025); AI Engineer for Data Scientists Associate (DataCamp, Aug 2025); Certified Data Analyst Associate (DataCamp, Aug 2025); Time Series Analysis (openHPI, Sep 2025).
- **Environmental & Marine Data:** Marine Data 4 Sustainable Oceans Workshop (Copernicus Marine Service / Mercator Ocean International, Apr 2025); Marine Data 4 Asia Workshop (Copernicus Marine Service / Mercator Ocean International, Jun 2025).
- **Forestry & Environment:** Policy Analysis and Forest Governance 101 (RECOFTC, Jun 2025); Advancing on REDD+ E-Academy (UNITAR / UN Environment, Oct 2024); Population Analysis Series: Tiger Density Analysis (Harimau Kita, Sep 2024).